

Time Stone

Technical Sheet — Participant Guide

*Predict the **Profit** for each transaction.*
Supervised Regression · Multivariate · Target: `Profit`

Data Space Club — ENSSEA

Quick Summary — Evaluation Weights

Metric	Weight	Interpretation
RMSE	40%	Penalizes large errors heavily
MAE	30%	Average absolute error (IDR)
MAPE	30%	Average percentage error

Contents

1 Files Provided **2**

2 The Target **2**

3 Column-by-Column Description **2**

3.1 Transaction Identifiers . . . 2
3.2 Product Information . . . 2
3.3 Time Information . . . 3
3.4 Customer & Channel . . . 3
3.5 Text-Derived Features (from `Item Name`) . . . 3
3.6 Derived Numeric Features . . . 3
3.7 Aggregation Features . . . 4
3.8 Operational Flags . . . 4
3.9 Noise (⚠ Beware) . . . 4

4 Important Notes for Participants **4**

4.1 Missing Values . . . 4
4.2 Redundant / Derived Features . . . 4
4.3 Categorical Encoding . . . 4
4.4 The Key Predictor . . . 5

5 Submission Format **5**

6 Evaluation Metrics **5**

7 Suggested Workflow **5**

8 Bonus **6**

1 Files Provided

File	Description
`train.csv`	Training data (70 %) — includes the target `Profit`
`test.csv`	Test data (30 %) — **without** the target `Profit`
`sample_submission.csv`	Submission template (format to follow)
`data_description.md`	This guide

2 The Target

Your Objective

Predict `Profit` (float, IDR) for each transaction in the test set.

`Profit = Item Price Amount − Material Cost`

Available in `train.csv` only. You must predict it for every row in `test.csv`.

3 Column-by-Column Description

3.1 Transaction Identifiers

#	Column	Type	Description
1	`Invoice Code`	str	Unique invoice ID (e.g. `SI24100331`). Multiple items can share the same invoice.
2	`Customer Code`	str	Anonymized customer ID (e.g. `P0535`). Some customers are walk-in.

3.2 Product Information

#	Column	Type	Description
3	`Item Name`	str	Product/service name (free text).
4	`Item Quantity`	float	Quantity sold (supports decimals, e.g. 4.4 m of banner).
5	`Item Price`	float	Unit price per item/meter (IDR).
6	`Item Price Amount`	float	Total price = `Item Quantity` × `Item Price` (IDR).
8	`Item Category`	str	Derived category: Banner, Sticker, ID & Lanyard, Stamp, Paper & Board, Photo & Poster, Apparel, Printed Docs, Souvenir, Other.
9	`Material Cost`	float	Cost of raw materials (IDR). **Key predictor of Profit.**

3.3 Time Information

#	Column	Type	Description
7	`Transaction At`	datetime	Timestamp (`YYYY-MM-DD HH:MM:SS`).
17	`hour`	int	Hour of the day (0–23).
18	`day_of_week`	int	Day of week (0=Mon, 6=Sun).
19	`day_of_month`	int	Day of month (1–31).
20	`month`	int	Month (1–12).
21	`quarter`	int	Quarter (1–4).
22	`week_of_year`	int	ISO week number (1–53).
23	`is_month_start`	int	1 if first days of month, else 0.
24	`is_month_end`	int	1 if last days of month, else 0.
15	`Is_Weekend`	int	1 if Saturday/Sunday, else 0.
16	`Is_Holiday`	int	1 if Indonesian public holiday, else 0.

3.4 Customer & Channel

#	Column	Type	Description
11	`Customer Type`	str	Individual, Small Business, Corporate, Walk-in.
12	`Order Channel`	str	WhatsApp, Instagram, Walk-in, Email, Website.
13	`Payment Method`	str	Cash, Transfer, QRIS, Credit.
14	`Discount (%)`	float	Discount applied. 0 if none.
40	`channel_prepaid`	int	1 if Payment Method $\in$ {Transfer, QRIS, Credit}.
41	`is_corporate`	int	1 if Customer Type == Corporate.

3.5 Text-Derived Features (from `Item Name`)

#	Column	Type	Description
25	`item_name_length`	int	Character length of `Item Name`.
26	`item_word_count`	int	Number of words in `Item Name`.
27	`has_size_keyword`	int	1 if name contains size markers (A4, F4, S/M/L, X2).
28	`has_number_in_name`	int	1 if name contains any digit.

3.6 Derived Numeric Features

#	Column	Type	Description
29	`price_per_unit_qty`	float	`Item Price / Item Quantity`.
30	`quantity_log`	float	`log1p(Item Quantity)` — log-transformed quantity.
31	`amount_log`	float	`log1p(Item Price Amount)` — log-transformed total.
32	`unit_price_bucket`	str	Price tier: cheap / mid / high / premium.

3.7 Aggregation Features

#	Column	Type	Description
33	`customer_visits`	int	Total transactions by this customer.
34	`customer_avg_amount`	float	Average `Item Price Amount` for this customer.
35	`customer_max_amount`	float	Maximum `Item Price Amount` for this customer.
36	`category_avg_price`	float	Average `Item Price` for this `Item Category`.
37	`category_count`	int	Number of transactions in this `Item Category`.

3.8 Operational Flags

#	Column	Type	Description
38	`is_bulk_order`	int	1 if `Item Quantity` is in the top 10 %.
39	`is_high_value`	int	1 if `Item Price Amount` is in the top 10 %.

3.9 Noise (⚠ Beware)

#	Column	Type	Description
42	`noise_feature`	float	Random Gaussian noise (mean=0, std=1). **No predictive value.** Tests your feature selection.

⚠ The Noise Trap

`noise_feature` is **purely random**. Including it in your model will hurt your score. Identify and drop it.

4 Important Notes for Participants

4.1 Missing Values

- ★ Some columns contain `NaN` values (5–15 % depending on column).
- ★ You must decide how to handle them: imputation, dropping, or modeling them as a separate category.

4.2 Redundant / Derived Features

Several columns are derived from others (e.g. `Is_Weekend` from `day_of_week`, `amount_log` from `Item Price Amount`).

⚠ Multicollinearity Risk

Not all features help. Some cause multicollinearity. Use feature selection (correlation analysis, VIF, tree-based importance).

4.3 Categorical Encoding

You must encode these categorical columns yourself:

- `Customer Code`, `Item Name`, `Item Category`, `Customer Type`, `Order Channel`, `Payment Method`, `unit_price_bucket`.
- **High-cardinality:** `Customer Code`, `Item Name` — consider target encoding or hashing.

4.4 The Key Predictor

Key Insight

`Material Cost` is strongly correlated with `Profit`. The naive formula `Profit = Item Price Amount − Material Cost` would give a perfect prediction — **but the challenge is to model Profit robustly, including cases where Material Cost is missing or noisy.**

5 Submission Format

Submit a CSV file named `submission.csv` with **exactly** this format:

```
index,predicted_profit
0,12345.67
1,89000.00
2,1500.50
...
```

- One row per test row (same order as `test.csv`).
- `index` = the row position in `test.csv` (0, 1, 2, ...).
- `predicted_profit` = your predicted `Profit` value (float, IDR).
- Do **not** include any header other than the one shown.

6 Evaluation Metrics

Your submission will be scored on **three metrics**:

Metric	Weight	Interpretation
RMSE (Root Mean Squared Error)	40%	Penalizes large errors heavily.
MAE (Mean Absolute Error)	30%	Average absolute error in IDR.
MAPE (Mean Absolute Percentage Error)	30%	Average percentage error.

Final Score Formula

$$\text{Final Score} = 0.40 \times \text{RMSE}_{\text{norm}} + 0.30 \times \text{MAE}_{\text{norm}} + 0.30 \times \text{MAPE}_{\text{norm}}$$

Lower is better.

7 Suggested Workflow

1. **EDA** — Explore distributions, missing patterns, correlations with `Profit`.

2. **Handle NaN** — Impute or drop. Document your strategy.
3. **Encode categoricals** — Choose: one-hot, label, target, or hashing.
4. **Feature selection** — Drop `noise_feature` and redundant columns. Use SHAP / correlation.
5. **Model** — Try: Linear Regression, Random Forest, XGBoost, LightGBM, CatBoost.
6. **Validate** — Use K-Fold CV (with `random_state=42` for reproducibility).
7. **Submit** — Match the exact submission format above.

8 Bonus

Bonus Points

- The first team to **identify and drop `noise_feature`** will receive a **+2 minute bonus** on their final presentation.
- Using advanced feature engineering (target encoding, interaction features) earns **extra points**.

★ Good luck! ★